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Cyanothiophene-Flanked Diketopyrrolopyrrole: A New Electron-Deficient Building Block for Unipolar n-Type Polymer Semiconductors

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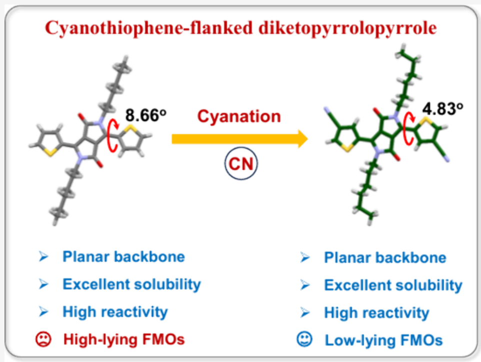
ABSTRACT: Developing high-performance n-type polymer semiconductors depends on the design and synthesis of strongly electron-deficient units with optimized physicochemical properties. Here, we report a novel diketopyrrolopyrrole derivative (**CTDPP**) by integrating electron-withdrawing cyano groups into the benchmark thiophene-flanked DPP (**TDPP**). The new **CTDPP** not only preserves the high backbone coplanarity of its parent structure, but also exhibits the deepest lowest unoccupied molecular orbital energy level among all DPP derivatives reported and a notably short $\pi-\pi$ stacking distance (~3.36 Å) in the single-crystal structure, rendering it an excellent building block for constructing high-performance n-type polymers. When incorporated into polymer backbones, the **CTDPP**-based polymers show unipolar n-type transport characteristics, with a maximum field-effect electron mobility exceeding 0.1 $cm^2\ V^{-1}\ s^{-1}$ and excellent n-type dopability even at low dopant loadings. Consequently, the polymers achieve remarkable n-type thermoelectric performance with a maximum electrical conductivity of 28.8 S cm^{-1} and power factor of 34.11 $\mu W\ m^{-1}\ K^{-2}$, thus leading to the highest figure-of-merit (ZT) of 0.08, which is among the highest values for unipolar n-type polymers. These results highlight the effectiveness of **CTDPP** as a versatile highly electron-deficient building block for high-performance n-type polymers and offer novel design principles for the development of DPP-based materials in organic electronics.



■ INTRODUCTION

Polymer semiconductors (PSs) are revolutionizing the landscape of organic electronics thanks to their intrinsic properties that enable lightweight, flexible, and large-area devices with low fabrication costs.[1−3] Applications such as organic field-effect transistors (OFETs), organic electrochemical transistors, and organic thermoelectrics (OTEs) are mainly powered by the advances in PS design and synthesis.[4−8] Over the past decade, tremendous progress has been made in p-type PSs, achieving high hole mobilities exceeding 10 $cm^2\ V^{-1}\ s^{-1}$ in OFETs and excellent figure-of-merit (ZT) surpassing 0.5 in OTEs, but the development of high-performance n-type polymers lags far behind.[9−14] As the integration of both high-performance p-type and n-type PSs is indispensable for enabling complementary logic circuits and high-efficiency thermoelectric modules, developing n-type polymers with comparable performance to their p-type counterparts becomes imperative.[14−16]

To achieve superior n-type performance, PSs should have sufficiently suppressed frontier molecular orbital (FMO) energy levels, including both the lowest unoccupied molecular orbital (LUMO) and the highest occupied molecular orbital (HOMO) energy levels, which will facilitate efficient electron injection/generation, stabilize electron transport, and concurrently suppress hole injection/generation.[4,17−23] One proven strategy to achieve this goal is to replace the typical donor−acceptor (D-A) polymer backbone with an acceptor−acceptor (A-A) one. This design effectively mitigates intramolecular charge transfer (ICT), thereby enabling polymers with simultaneously deep LUMO and HOMO levels.[24−27] Therefore, the success of developing high-performance n-type polymers critically relies on the availability of strongly electron-deficient building blocks, which remains a significant bottleneck due to the limited number of such units that combine high electron affinity with favorable physicochemical properties.[28−31] To address this challenge, extensive researches have been dedicated to introducing strong electron-withdrawing groups and atoms, such as imide,[32−34] amide,[4,35] boron−nitrogen coordination bonds,[36] halogens,[37] and cyano groups,[38] into π-conjugated aromatic units (e.g., polycyclic

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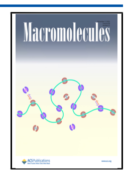


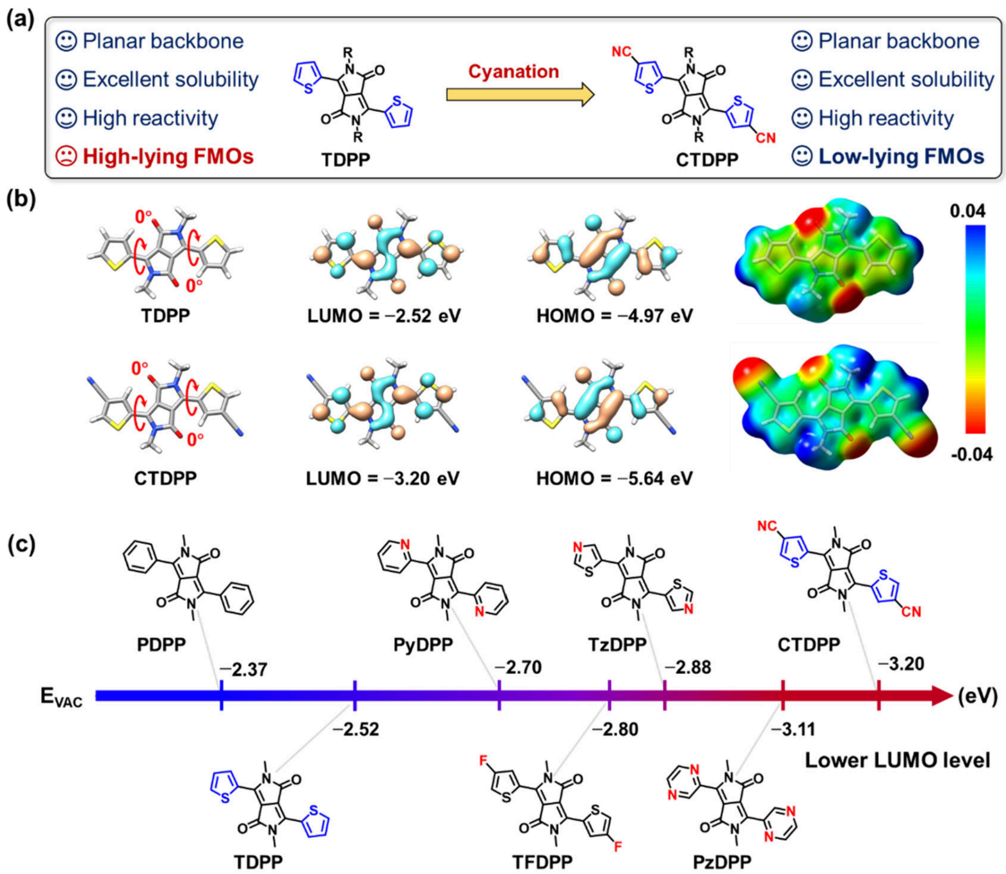


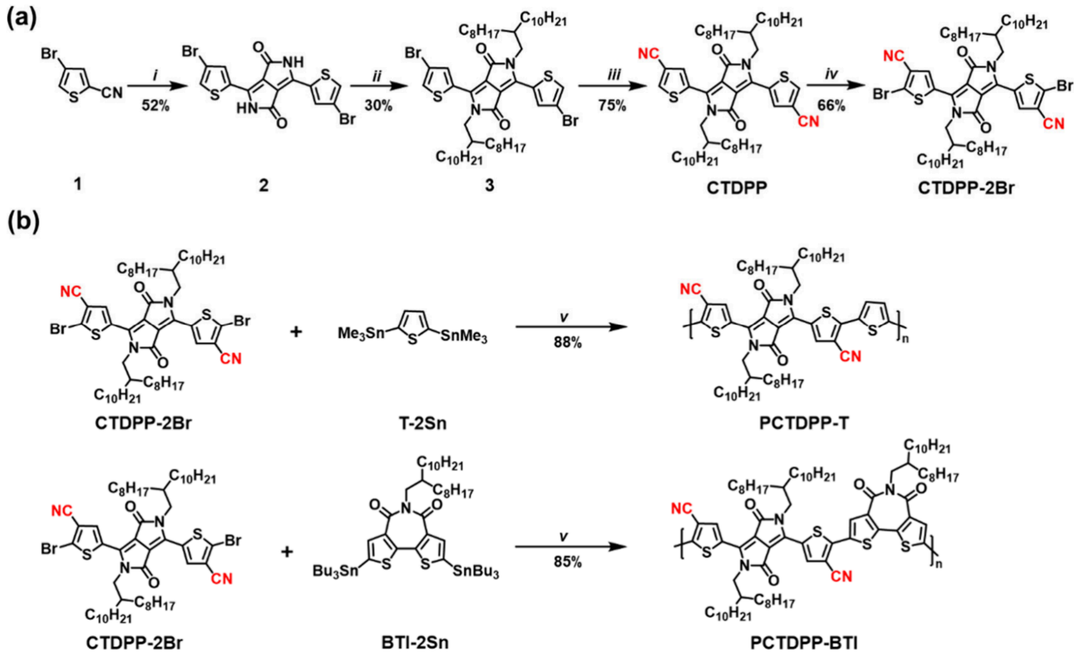
Figure 1. (a) Molecular design strategy for cyano-functionalized building block **CTDPP**. (b) DFT-optimized geometries and ESP maps of **CTDPP** and its parent **TDPP** unit, calculated at the B3LYP/6−31G** level. (c) Evolution of LUMO energy levels of **CTDPP** vs those of previously reported diketopyrrolopyrrole derivatives.

aromatic hydrocarbons), offering an efficient tactic to construct highly electron-deficient cores.[39]

Among various electron-deficient building blocks, thiophene-flanked diketopyrrolopyrrole (**TDPP**) has emerged as a highly popular and versatile candidate for constructing PSs due to its planar π-conjugated framework, strong $\pi-\pi$ interactions, excellent solubility imparted by *N*-alkylation, and high reactivity during polymerization.[40,41] However, the presence of two electron-rich thiophene moieties in **TDPP** raises the FMO energy levels of the resulting **TDPP**-based PSs, typically leading to p-type or ambipolar transport characteristics.[42,43] Indeed, this limitation has attracted considerable attention, recently,[44−46] prompting the design of various electron-deficient **TDPP** derivatives through the strategic incorporation of more electron-withdrawing heteroarenes, such as pyridine-substituted DPP,[47] thiazole-flanked DPP,[48] and pyrazine-flanked **DPP** (**PzDPP**).[49] Despite these advances, the performance of the resulting unipolar n-type polymers still lags behind that of their p-type counterparts. For example, the thiazole-flanked DPP-based polymer **PTz-5-DPP** exhibits impressively low-lying LUMO/HOMO levels of −3.85/−5.70 eV and a highly coplanar backbone conformation, but achieves a low electron mobility (μ_e) of 0.03 $cm^2\ V^{-1}\ s^{-1}$ in OFETs and only moderate n-doped electrical conductivity (σ) of 8.4 S cm^{-1} in OTEs, both of which remain well below those of benchmark p-type materials.[48] The scenario highlights an urgent need to explore new electron-deficient building blocks with optimized molecular geometries and electronic properties for developing high-performance n-type PSs.[42]

Cyano group has been widely employed as a simple yet potent electron-withdrawing substituent that markedly enhances electron deficiency in conjugated systems, particularly when combined with other withdrawing motifs.[38] Compared with alternative electron-withdrawing motifs, the cyano substitution offers several distinctive advantages: (i) a stronger electron-withdrawing effect, which effectively lowers the LUMO energy level and thereby reduces electron injection barriers; (ii) enhanced π-electron delocalization arising from sp-hybridization, which contributes to backbone planarity and charge delocalization; and (iii) a larger dipole moment, which strengthens intermolecular interactions and promotes favorable molecular packing. Cyano functionalization has thus led to the development of a variety of novel building blocks, such as cyano-functionalized perylene/naphthalene diimide,[50,51] dicyano bithiophene imide (**BTI**),[27] and cyanated imide-functionalized fluorenone,[52] which exhibit deeper-lying LUMO energy levels (often >0.3 eV lower than their noncyanated analogues) while retaining good solubility. Therefore, PSs based on these cyano-functionalized building blocks demonstrated great potential in realizing unipolar n-type transport characteristics. For example, polymers based on fused cyanated bithiophene imide dimer showed low LUMO levels down to − 3.90 eV and unipolar μ_es exceeding 0.10 $cm^2\ V^{-1}\ s^{-1}$ in OFETs.[53] Building on this rationale, it is therefore expected that incorporating cyano groups into **TDPP** could afford novel acceptor units

Scheme 1. Synthetic Route to (a) the Dibrominated Monomer CTDPP-Br and (b) the Corresponding Polymers PCTDPP-T and PCTDPP-BTI[a]



[a]Reagents and conditions: (*i*) diisopropyl succinate, Na, 2-methyl-2-butanol, 105 °C; (*ii*) K_2CO_3, 1-bromodecane, DMF, 100 °C; (*iii*) $Zn(CN)_2$, $Pd_2(dba)_3$, dppf, 1,4-dioxane, 100 °C; (*iv*) LDA, $BrCl_2CHCHCl_2Br$, THF, −78 °C; (*v*) $Pd_2(dba)_3$, P(*o*-tol)$_3$, CuI, toluene, microwave, 140 °C.

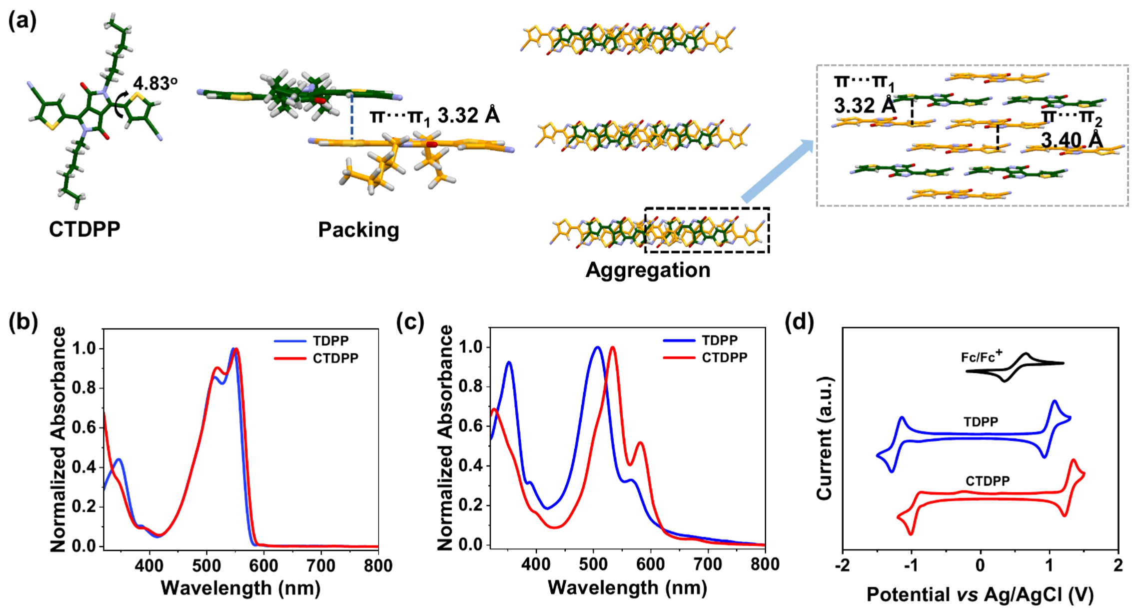


Figure 2. (a) Single-crystal structures of model compound **CTDPP(C6)**. Normalized UV−Vis absorption spectra of **TDPP** and **CTDPP** were measured (a) in diluted chloroform solutions and (b) as thin films. (c) Cyclic voltammograms of **TDPP** and **CTDPP** recorded in acetonitrile using the Fc/Fc^+ redox couple as the external standard.

with significantly lowered FMO energy levels while maintaining an ordered molecular arrangement and good solubility.

In this context, we report a novel highly electron-deficient building block, 4-cyanothiophene-flanked diketopyrrolopyrrole **CTDPP** (Figure 1a), developed via a cyano-functionalization strategy applied to the **TDPP** framework. Density functional theory (DFT) calculations reveal that cyanation in **CTDPP** significantly lowers the FMO energy levels compared to its parent **TDPP** (Figure 1b), with **CTDPP** exhibiting the lowest LUMO level (−3.22 eV) among the widely used **DPP** derivatives reported to date (Figure 1c).[49] Moreover, **CTDPP** retains the high coplanarity of the **TDPP** core, making it a promising building block for the construction of high-performance n-type PSs. Guided by these theoretical insights, we successfully synthesized the dibrominated monomer **CTDPP-2Br** and subsequently copolymerized it with distannylated thiophene (**T-2Sn**) and bithiophene imide (**BTI-2Sn**) to afford D-A polymer **PCTDPP-T** and A-A polymer **PCTDPP-BTI**, respectively. Both polymers exhibit good solubility, deep-lying FMO energy levels, and decent backbone coplanarity, leading to unipolar n-type transport characteristics with a maximum μ_e of 0.11 cm^2 V^{-1} s^{-1} in OFETs and high thermoelectric ZT value of 0.08 in OTEs. These results are superior to those of most TDPP-based polymers, which typically show p-type behavior, demonstrating that **CTDPP** is a highly versatile and promising electron-deficient building block for the design of high-performance n-type PSs.

■ RESULTS AND DISCUSSION

Scheme 1a illustrates the synthetic route to **CTDPP** and its dibrominated monomer **CTDPP-2Br**, following a succinate-based protocol.[41] Briefly, commercially available 4-bromothiophene-2-carbonitrile (compound **1**) was first condensed with diisopropyl succinate in 2-methyl-2-butanol using *tert*-BuOK as the base to afford brominated compound **2** in 52% yield. Subsequent *N*-alkylation using 2-octyldodecyl chains followed by cyanation with zinc cyanide yielded target cyanated **CTDPP**. All of the intermediates and the final monomer **CTDPP-2Br** were fully characterized by ^{1}H NMR and ^{13}C NMR spectroscopy, as detailed in the Supporting Information. To shed light on the molecular geometry and solid-state packing of the new cyanated building block, **CTDPP(C6)** bearing linear *n*-hexyl side chains on the amide groups was synthesized as a model compound (Scheme S1). Single crystals of **CTDPP(C6)** suitable for X-ray diffraction were obtained by slow evaporation from a chloroform/methanol mixture.

As displayed in Figure 2a, **CTDPP(C6)** features a smaller dihedral angle of 4.83° between the central **DPP** core and the neighboring 4-cyanothiophene compared to that of **TDPP-(C6)**, indicating a more planar framework for **CTDPP**, as predicted by DFT calculations (see Figure S1). More importantly, in terms of molecular self-assembly, **TDPP(C6)** adopts a herringbone packing motif with a $\pi-\pi$ stacking distance of 3.50 Å (Figure S2),[54] whereas **CTDPP(C6)** features a one-dimensional slipped-stacking structure with a reduced $\pi-\pi$ stacking distance of ~3.36 Å.[55] This reduced distance in **CTDPP(C6)** suggests a more compact packing and enhanced intermolecular interaction induced by cyano functionalization, likely due to stronger electrostatic interactions as anticipated by the DFT-calculated electrostatic potential (ESP) maps (Figure 1b). Specifically, the ESP maps show a pronounced electron-deficient region around the cyano groups and an electron-rich area over the conjugated backbone, highlighting the increased polarity and the potential for dipole−dipole interactions that promote a tighter molecular packing.[56] Such improved coplanarity combined with a more compact packing motif is expected to facilitate efficient charge carrier transport in the **CTDPP**-based PSs.

Interestingly, ultraviolet−visible (UV−vis) absorption spectra of both **CTDPP** and **TDPP** uncover that cyano functionalization modulates not only the structural properties but also the optoelectronic characteristics of the molecules. In chloroform solution, both exhibit strong absorption bands in the range of 400−600 nm, associated with local $\pi-\pi^*$ electronic excitations,[57] with maximum absorption wavelengths (λ_{max}^{soln}s) at 546 nm for **CTDPP** and 552 nm for **TDPP** (Figure 2b). Upon casting into thin films (Figure 2c), **CTDPP** exhibits a larger bathochromic shift of 30 nm compared to a 20 nm shift observed for **TDPP**, likely due to enhanced intermolecular aggregation promoted by cyano functionalization.[56,58] To probe how the structural modification affects FMO energy levels, cyclic voltammetry (CV) measurements were carried out on both compounds. As shown in Figure 2d, incorporation of the electron-withdrawing CN groups into the TDPP core lowers the reduction potential and increases the oxidation potential, indicating an enhanced n-type character. On the basis of the reduction/oxidation onsets, the LUMO/HOMO energy levels (E_{LUMO}/E_{HOMO}s) are derived to be −3.15/−5.26 and −3.42/−5.52 eV for **TDPP** and **CTDPP**, respectively. Apparently, cyano functionalization effectively lowers the FMO levels by ~0.3 eV, which is in good accordance with the trend found in the DFT calculations. The calculated electrochemical bandgaps (E_g^{CV}s) are 2.11 eV for **TDPP** and 2.10 eV for **CTDPP**, which aligns well with the optical bandgaps (2.04 eV for **TDPP** vs 2.01 eV for **CTDPP**) estimated from UV−vis absorption onsets. Furthermore, **CTDPP** demonstrates a superior thermal stability compared to **TDPP**, as evidenced by its higher decomposition temperature (T_d = 370 °C vs 350 °C) measured by thermogravimetric analysis (TGA) (Figure S3), and elevated melting transition temperature (T_m = 130 °C vs 80 °C) revealed by differential scanning calorimetry (DSC) (Figure S4). These combined structural, optoelectronic, and thermal properties highlight the significant impact of cyano functionalization in enhancing the performance potential of CTDPP-based materials.

Based on these insights, we aimed at incorporating the new **CTDPP** building block into polymer backbones; however, initial attempts to brominate **CTDPP** via conventional electrophilic substitution methods (e.g., using *N*-bromosuccinimide or elemental bromine) proved unsuccessful, as it failed to yield the desired dibrominated monomer. Consequently, lithiation of the α-position hydrogens on the cyanothiophene units in **CTDPP** was carried out using the strong base and weak nucleophile lithium diisopropylamide (LDA) at −78 °C, followed by treatment with 1,2-dibromotetrachloroethane to afford the dibrominated monomer **CTDPP-2Br** (Scheme 1a). This monomer was subsequently copolymerized with distannylated thiophene (T-2Sn) to yield D-A polymer **PCTDPP-T** (Scheme 1b). In addition, the A-A polymer **PCTDPP-BTI** was also synthesized by copolymerizing **CTDPP-2Br** with distannylated **BTI-2Sn**, benefiting from the small steric hindrance of both the **CTDPP** and **BTI** units. After polycondensation, sequential Soxhlet extraction was performed using methanol, acetone, *n*-hexane, and dichloromethane to remove oligomers and impurities, with the final chloroform fractions collected as the purified polymers for further characterization and device fabrication. High-temperature (150 °C) gel permeation chromatography (GPC) measurements revealed number-average molecular weights (M_ns)/polydispersity indeces (Đs) of 42.0 kDa/2.5 for **PCTDPP-T** and 87.4 kDa/2.3 for **PCTDPP-BTI** (Figures S5 and S6). TGA demonstrated excellent thermal stability for both polymers, with decomposition temperatures exceeding 350 °C. Additionally, no distinctive thermal transitions were observed for either polymer in the DSC curves within the range of 30~300 °C, further indicating high thermal robustness.

To gain a first perception on how cyano functionalization and different comonomer units affect the structural and electronic properties of the newly synthesized polymers, DFT calculations were performed on trimeric models of these polymers at the B3LYP-D3/6−31G** level using the Gaussian 16 software. As shown in Figure S7, the introduction of cyano groups induces a certain degree of steric hindrance between the **CTDPP** unit and the adjacent comonomers, resulting in

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Table 1. Summary of Molecular Weights, Optical and Electrochemical Properties of Building Blocks TDPP and CTDPP, and Polymers PCTDPP-T and PCTDPP-BTI

material	M_n (kDa)/$Đ$	$\lambda_{max}^{soln\,a}$ (nm)	$\lambda_{max}^{film\,b}$ (nm)	$E_g^{opt\,c}$ (nm)	$E_{LUMO}^{\ d}$ (eV)	$E_{HOMO}^{\ d}$ (eV)
TDPP		546	566	2.04	−3.15	−5.26
CTDPP		552	582	2.01	−3.42	−5.52
PCTDPP-T	42.0/2.5	760	800	1.29	−3.84	−5.77
PCTDPP-BTI	87.4/2.3	732	756	1.44	−3.90	−5.87

[a]Measured in dilute chloroform solutions (10^{-5} M); [b]Measured as thin films spin-coated from chloroform solutions; [c]Estimated from the absorption onset of thin-film UV−vis spectra; [d]Calculated from the onset potentials of thin-film cyclic voltammograms recorded in 0.1 M tetra(*n*-butyl)ammonium hexafluorophosphate in acetonitrile, using Fc/Fc$^+$ as the external reference.

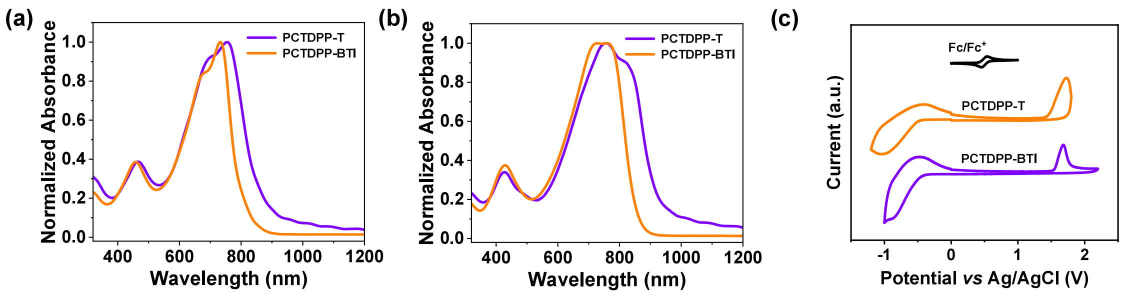

Figure 3. Normalized UV−vis−NIR absorption spectra of polymers **PCTDPP-T** and **PCTDPP-BTI** (a) in diluted chloroform solutions and (b) as thin films. (c) Cyclic voltammograms of **PCTDPP-T** and **PCTDPP-BTI** using the Fc/Fc$^+$ redox couple as the external standard.

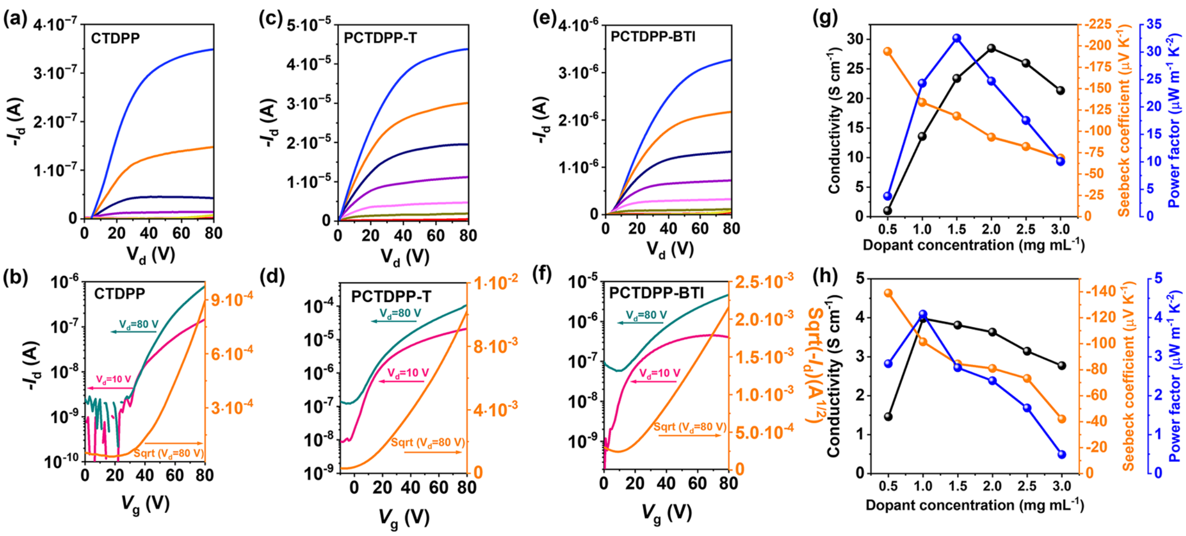

Figure 4. (a−c) Output and (d−f) transfer characteristics of top-gate/bottom-contact OFETs based on (a, d) the **CTDPP** small molecule and the polymers (b, e) **PCTDPP-T** and (c, f) **PCTDPP-BTI**. Device dimensions: L = 10 μm, W = 5 mm. (g, h) Thermoelectric performance parameters, including electrical conductivities, Seebeck coefficients, and power factors for (g) **PCTDPP-T** and (h) **PCTDPP-BTI**, measured at different JLBI dopant concentrations in the range of 1−3 mg mL^{-1}.

dihedral angles of approximately 17° for **PCTDPP-T** and 9° for **PCTDPP-BTI**. Despite this torsion, the polymer backbones remain sufficiently planar to allow effective delocalization of both HOMO and LUMO energy levels along the π-conjugated backbones. Notably, **PCTDPP-BTI** exhibits a more pronounced delocalization of the FMO energy levels along the backbone, which is attributed to its better coplanarity provided by the rigid and highly conjugated A-A architecture of **BTI** and **CTDPP** units. The resulting improved π-conjugation leads to lower-lying FMO energy levels, with predicted E_{LUMO}/E_{HOMO} values of −3.73/−5.45 eV for **PCTDPP-BTI**, approximately 0.13 eV deeper than those of the D-A type **PCTDPP-T** polymer (−3.61/−5.32 eV).

Importantly, these theoretically predicted trends are consistent with the experimental values obtained from the ultraviolet − visible − near-infrared (UV−vis−NIR) absorption spectra of the CTDPP-based polymers, with the corresponding parameters summarized in Table 1. Both polymers display the typical dual-peak absorption profile of TDPP-based materials in dilute chloroform solution (Figure 3a),[57] featuring a weak absorption band in the 400−500 nm range and a strong shoulder-shaped band extending into the near-infrared region (700−900 nm). Compared to **PCTDPP-T**, **PCTDPP-BTI** exhibits a significant blue shift of ∼30 nm in its absorption maximum, which is likely attributed to the diminished ICT character associated with the A-A backbone configuration of **PCTDPP-BTI**. To study the aggregation

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behavior, we performed variable-temperature UV−Vis absorption spectroscopy (Figure S8). Both polymers exhibit a gradual decrease in absorption intensity with increasing temperature, indicating strong aggregation in solution. These results indicate that the two polymers possess comparable aggregation tendencies in solution. Upon transitioning from solution to the film state, a distinctive red-shift of 40 and 24 nm was observed in the low-energy absorption peaks of **PCTDPP-T** and **PCTDPP-BTI**, respectively (Figure 3b). The larger bathochromic shift in **PCTDPP-T** suggests a more ordered packing of polymer chains in the solid state, as further confirmed by film microstructure analyses (*vide infra*). Based on the absorption onsets of the polymer films, the optical bandgaps (E_g^{opt}s) of **PCTDPP-T** and **PCTDPP-BTI** were estimated to be 1.29 and 1.44 eV, respectively. This increase in E_g^{opt} for the A-A backbone polymer **PCTDPP-BTI**, which is 0.15 eV higher than the D-A polymer **PCTDPP-T**, further highlights the role of comonomer units in controlling ICT. To better understand this fine modulation of the FMO energy levels, HOMO and LUMO energy levels were determined by CV (Figure 3c), with the results summarized in Table 1. The CV curves reveal that **PCTDPP-T** exhibits E_{LUMO}/E_{HOMO} values of −3.84/−5.77 eV, both slightly higher than those of **PDPPCN-BTI** (−3.90/−5.87 eV). On the basis of these FMO energy levels, the electrochemical gap of **PCTDPP-T** was calculated to be 1.93 eV, slightly narrower than that (1.97 eV) of **PDPPCN-BTI**, which is in excellent agreement with the trend derived from the UV−vis−NIR absorption results.

This consistent correlation between theoretical modeling, electrochemical and optical data confirms the clear influence of polymer backbone architecture and cyano functionalization on the resulting structural and electronic properties, which naturally raises questions about how these molecular-level modifications translate into charge transport behavior. To address this, the charge-transport characteristics of **CTDPP** and its polymers were evaluated by fabricating OFETs with a top-gate/bottom-contact device architecture (see Supporting Information for detailed fabrication procedures). The output and transfer curves are depicted in Figure 4a−f, with the key performance parameters summarized in Table 2 and Table S1.

Table 2. Top-Gate/Bottom-Contact OFET and OTE Performance Parameters of the CTDPP-Based Polymer Semiconductors[a]

polymer	$\mu_{e,sat}$ ($cm^2\ V^{-1}\ s^{-1}$)	σ ($S\ cm^{-1}$)	S ($\mu V\ K^{-1}$)	PF ($\mu W\ m^{-1}\ K^{-2}$)
PCTDPP-T	0.113	28.87	−120.1	34.11
	(0.101 ± 0.0012)	(28.49 ± 0.20)	(−117.9 ± 1.58)	(32.53 ± 0.95)
PCTDPP-BTI	0.013	4.13	−103.8	4.44
	(0.010 ± 0.003)	(3.98 ± 0.14)	(−101.4 ± 2.36)	(4.09 ± 0.19)

[a]Maximum values are included, with average values from five independent devices shown in parentheses.

It is noteworthy that the TDPP-based polymers typically exhibit ambipolar or even p-type behavior, primarily due to the presence of two electron-donating thiophene moieties within the TDPP unit and the direct involvement of nitrogen lone pairs from the amide groups in the polymer backbone.[44,45] Nevertheless, both small molecule **CTDPP** and its corresponding polymers investigated in this work exhibit unipolar n-type transport characteristics, as evidenced in Figures 4d−f. These materials show low off-currents (∼10^{-7}−10^{-8} A) and moderate on/off current ratios ($I_{on}/I_{off} \approx 10^3$), presumably due to the effective stabilization of both LUMO and HOMO levels enabled by the incorporation of strong electron-withdrawing cyano groups. Specifically, the small molecule **CTDPP** demonstrated unexpected unipolar n-type transport behavior with a μ_e of 0.0016 $cm^2\ V^{-1}\ s^{-1}$, contrasting sharply with the negligible n-type OFET performance observed for its noncyanated counterpart **TDPP**. These results underscore the effectiveness of cyano functionalization in promoting efficient electron transport behavior, a fact further supported by the pronounced unipolar n-type transport observed in polymers **PCTDPP-T** and **PCTDPP-BTI**. Upon systematic optimization of the OFET fabrication conditions, polymers **PCTDPP-T** and **PCTDPP-BTI** achieved maximum μ_e values of 0.11 and 0.013 $cm^2\ V^{-1}\ s^{-1}$, respectively, with threshold voltages (V_{th}s) of 27 and 30 V. The superior μ_e of **PCTDPP-T** is attributed to its favorable polymer chain orientation and enhanced film crystallinity, as elaborated below. Note that the planar core and short π-π distance observed in the **CTDPP** single-crystal structure do not directly translate into the high electron mobility in its polymers. When **CTDPP** is incorporated into copolymers, the cyano substituents introduce steric repulsion with adjacent comonomers. This steric effect induces non-negligible backbone twists, as evidenced by average dihedral angles of ∼ 17° for **PCTDPP-T** and ∼ 9° for **PCTDPP-BTI**, thereby disrupting backbone planarity. The resulting distortion weakens intermolecular interactions, reduces the film crystallinity, and ultimately compromises charge transport. To assess operational stability of **PCTDPP-T**- and **PCTDPP-BTI**-based OFETs, unencapsulated devices were aged in ambient air (relative humidity: 40−60%). As shown in Figure S9, the electron mobility of both devices decayed to <10% of the initial value within 7 h. These degradations are attributed to the relatively high LUMO levels of both polymers, which increase susceptibility to oxygen and moisture.

Recognizing the crucial role of molecular doping in enhancing electrical conductivity ($\sigma = \mu ne$, where n is the charge carrier concentration and e is the elementary charge),[14] the low-lying LUMO energy levels and intrinsic n-type character of the **CTDPP**-based polymers motivated a comprehensive investigation of their n-doping capability. From this perspective, UV−vis−NIR absorption spectroscopy was initially employed to monitor n-polaron formation in polymer films upon increasing the dopant concentrations. Both polymers were n-doped using a sequential doping protocol assisted by transition metal catalyst.[59] The julolidine-functionalized benzimidazoline-based molecular dopant JLBI was used for its excellent solution processability, strong reducing capacity, and minimal microstructure disruption during n-doping.[60,61] As shown in Figure S10, doping with JLBI led to significant bleaching of the neutral π−π* absorption bands in both polymer films, accompanied by the emergence of two new sub-bandgap polaronic absorption bands in the visible/near-infrared region, indicative of an efficient n-doping process. To quantify this effect, the doping level was estimated using the polaron/π−π* peak intensity ratio. Interestingly, a higher polaron/π−π* ratio was observed for **PCTDPP-T** compared to that for **PCTDPP-BTI**, indicating a greater concentration of charged polymer chains (Figure S11). This trend is further supported by electron spin resonance (ESR) measurements for the JLBI-doped polymer films (Figure S12). As expected, no

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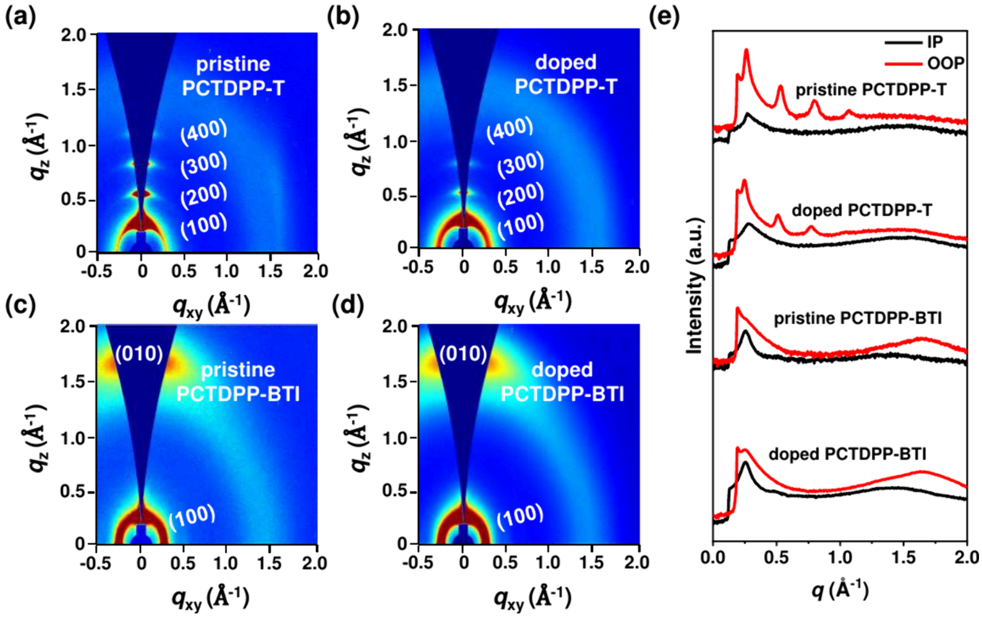

Figure 5. 2D-GIWAXS patterns of (a, b) **PCTDPP-T** and (c, d) **PCTDPP-BTI** films in their (a, c) pristine and (b, d) doped states. (e) Corresponding in-plane and out-of-plane line-cut profiles extracted from the 2D-GIWAXS images.

radical signals were observed for pristine films, while doped films showed increasing signal intensity with higher dopant loading consistent with a growing population of charge carriers. Curiously, JLBI-doped **PCTDPP-T** films exhibited slightly stronger ESR signals compared to those of **PCTDPP-BTI**, thereby reflecting more efficient charge generation in **PCTDPP-T**. This finding is somewhat intriguing given that the slightly higher-lying LUMO level of **PCTDPP-T** (−3.84 eV) relative to that of **PCTDPP-BTI** (−3.90 eV) would typically be expected to hinder electron transfer from the activated molecular dopant. This suggests that factors beyond energetic alignment, such as polymer−dopant miscibility and microstructural compatibility, play a critical role in determining the overall doping efficiency, as further discussed below.

Consequently, n-doped **PCTDPP-T** films exhibited a maximum σ of 28.8 S cm^{-1} at a JLBI dopant concentration of 2 mg mL^{-1}, which is substantially higher than that (4.2 S cm^{-1}) achieved for n-doped **PCTDPP-BTI** (Figure 4g-h). Such enhanced σ in **PCTDPP-T** can be attributed to the synergistic effects of its intrinsically higher μ_e and more efficient molecular doping. To quantify thermoelectric performance, Seebeck coefficients (S) of JLBI-doped films were determined by measuring the thermovoltage generated across the samples under applied temperature gradients from 0 to 2.8 K in 0.7 K increments (Figures S14and S15). As shown in Figure 4g,h, the doped films exhibited negative S values, characteristic of negative polarons proper for unipolar n-type materials. Increasing JLBI concentration from 0.5 to 3 mg mL^{-1} resulted in a gradual decrease in $|S|$, reflecting the typical inverse correlation between S and n.[62,63] Accordingly, heavily doped films with high σ values showed significantly low S and *vice versa*. Consequently, the optimal n-doping conditions that achieved the most balanced combination of σ and S were found at JLBI concentrations of 1.5 mg mL^{-1} for **PCTDPP-T** and 1.0 mg mL^{-1} for **PCTDPP-BTI**, yielding maximum power factors ($PF = S\sigma^2$) of 34.11 and 4.44 μW m^{-1} K^{-2}, respectively. For benchmarking against the widely used n-dopant N-DMBI, we performed parallel sequential-doping experiments under identical processing conditions (Figure S16). N-DMBI-doped **PCTDPP-T** and **PCTDPP-BTI** films yielded maximum power factors of 11.0 and 2.4 μW m^{-1} K^{-2}, respectively, substantially lower than those achieved with JLBI. These merits are significantly lower than those achieved using JLBI as the dopant, clearly demonstrating the superior doping efficiency of JLBI for our polymers.

Such a high PF achieved for JLBI-doped **PCTDPP-T** films motivated us to evaluate the thermal conductivity (κ) at room temperature using the transient hot-wire method,[64,65] given its critical role in determining the thermoelectric figure of merit ($ZT = \sigma S^2 T/\kappa$). In this context, the pristine **PCTDPP-T** film displayed a low κ of 0.139 ± 0.005 W m^{-1} K^{-1} at 300 K, highlighting its excellent thermal insulation compared to typical n-type organic semiconductors, which typically exhibit values exceeding 0.2 W m^{-1} K^{-1}.[62,63,66] Interestingly, κ decreased further with increasing the JLBI concentration due to enhanced phonon scattering by free carriers and dopant-induced disruptions in molecular packing,[67−69] reaching a minimum of 0.110 ± 0.004 W m^{-1} K^{-1} at the highest dopant loadings (3 mg mL^{-1}). Notably, the most balanced combination of PF (34.11 μW m^{-1} K^{-2}) and κ (0.124 W m^{-1} K^{-1}) was obtained at a JLBI dopant concentration of 1.5 mg mL^{-1}, resulting in a high ZT value of 0.08 at room temperature (see Figure S17). To our knowledge, this value is among the highest ZT reported for purely n-type organic thermoelectrics,[70−76] firmly establishing **CTDPP** as a promising design motif for the next-generation plastic electronics. To assess the stability, we tested PCTDPP-T- and PCTDPP-BTI-based OTEs under ambient air (relative humidity: 40−60%) without encapsulation. JLBI-doped films showed a rapid decline in conductivity with unstable Seebeck signals, which prevented reliable curves. These fast degradations can be ascribed to the relatively high LUMO levels of the CTDPP-based polymers.

To elucidate the relationship between material structures and device performance, the morphology of polymer thin films was first characterized via atomic force microscopy (AFM). As

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shown in Figure S18, both pristine **PCTDPP-T** and **PCTDPP-BTI** films exhibit smooth and uniform surfaces with low root-mean-square (RMS) roughness values below 1 nm, indicating minimal surface irregularities and suppressed grain boundaries. Upon n-doping with JLBI, no obvious surface aggregation or phase separation was observed, as evidenced by the nearly unchanged RMS roughness. This suggests effective n-dopant diffusion into the polymer matrices, a critical factor for maintaining film integrity and electronic performance.[53]

To further probe the film nanostructures, we employed 2D grazing-incidence wide-angle X-ray scattering (2D-GIWAXS) to explore the intermolecular packing and orientation of both pristine and doped films. In this context, 2D-GIWAXS images revealed clear differences in the molecular packing of two pristine polymers (Figure 5a,b). It is found that **PCTDPP-T** adopts a predominantly edge-on orientation, as evidenced by strong lamellar diffraction peaks progressing up to the (400) order in the out-of-plane (OOP) direction and a pronounced $\pi-\pi$ stacking peak (010) at q_{xy} = 1.60 Å^{-1} in the in-plane (IP) direction. In contrast, **PCTDPP-BTI** features a face-on packing motif, with a prominent (010) $\pi-\pi$ stacking diffraction peak in the OOP direction at q_z = 1.68 Å^{-1}, together with a (100) lamellar peak in the IP direction at q_{xy} = 0.25 Å^{-1}. In addition, film crystallinity analysis based on the Scherrer equation revealed that **PCTDPP-T** exhibits a significantly larger crystalline coherence length (CCL) along the lamellar direction compared to **PCTDPP-BTI** (271 Å vs 104 Å) (see Table S2). This, together with its more ordered stack extending up to the fourth order, further supports the superior n-type performance observed in **PCTDPP-T**-based OFETs. Interestingly, scrutiny of the line-cut profiles uncovers a slightly larger $\pi-\pi$ stacking distance for **PCTDPP-T** (3.97 Å) compared to that for **PCTDPP-BTI** (3.74 Å), likely due to the more twisted backbone of **PCTDPP-T** that partially hinders $\pi-\pi$ overlapping between adjacent polymer chains. This observation is somewhat intriguing considering the lower μ_e measured for **PCTDPP-BTI**, whose more planar and closely packed structure should, in principle, promote both intrachain electron delocalization and interchain electron hopping. However, it is important to note that the predominant face-on orientation adopted by this polymer is a $\pi-\pi$ stacking direction that is orthogonal to the lateral charge transport pathway in our device architecture, thereby limiting charge transport between electrodes. Based on these findings, the higher μ_e observed in **PCTDPP-T**-based OFETs can be mainly attributed to its high film crystallinity and predominantly edge-on orientation, which promotes efficient in-plane charge transport consistent with the lateral charge transport pathway of the top-gate/bottom-contact device architecture.[77]

Given that incorporation of molecular dopants often alters supramolecular ordering while diffusing through the polymer matrix,[68,73,78] we investigated how n-doping influences the microstructure of our materials by analyzing the 2D-GIWAXS patterns. Interestingly, it was found that although both polymers retained their supramolecular orientation during the doping process, the $\pi-\pi$ stacking distance of **PCTDPP-T** markedly increased to 4.11 Å, a value that clearly exceeds the 3.79 Å stacking distance observed in the doped films of **PCTDPP-BTI**. We tentatively attribute this observation to partial interactions between the dopant molecules and the polymer backbone with a more pronounced effect for **PCTDPP-T** in accordance with its higher n-doping efficiency. On the other hand, the more densely ordered supramolecular arrangement of **PCTDPP-BTI** may limit dopant permeation to the polymer backbone, leading to less efficient electron transfer processes despite its LUMO energy level being comparable to that of **PCTDPP-T**. As a consequence, the integral areas of both the (010) and the (100) stacking peaks exhibit only minor variations. In fact, the doped **PCTDPP-T** maintained a large $CCL_{z,010}$ of 182.42 Å, still much larger than that of **PCTDPP-BTI** (104.72 Å). These results indicate that **PCTDPP-T** preserves a high degree of crystallinity while allowing for an effective n-doping process, key attributes that contribute to its higher electrical σ and charge carrier concentrations, ultimately leading to superior thermoelectric performance.

■ CONCLUSION

To summarize, a structurally novel and highly electron-deficient building block **CTDPP** was designed and synthesized via elaborately integrating cyano functionalities into the classical amide-functionalized **TDPP** unit. The newly developed **CTDPP** not only maintains the excellent properties of its parent **TDPP** (i.e., good solubility, high degree of backbone coplanarity, and high reactivity) but also features a small $\pi-\pi$ stacking distance and deep-positioned FMO levels. Consequently, the OFETs with building block **CTDPP** as the active layer exhibit unipolar n-type character with an encouraging μ_e of 0.0016 cm^2 V^{-1} s^{-1}, substantially outperforming the negligible OFET characteristics achieved for its parent **TDPP**. Copolymerizing **CTDPP** with donor counit thiophene and acceptor counit bithiophene imide enables the resultant copolymers achieving unipolar n-type performance, with the champion μ_e of >0.1 cm^2 V^{-1} s^{-1} achieved for polymer **PCTDPP-T**. Owing to their low-lying FMO energy levels and good electron transport characteristics, the **CTDPP**-based polymers can be efficiently n-doped using JLBI as the molecular dopant and deliver a sizable σ of 28.9 S cm^{-1} and an excellent *ZT* of 0.08. These findings demonstrate that cyanation of amide-functionalized arenes is a powerful approach for constructing strongly electron-deficient units and offer valuable design principles for developing next-generation DPP-based unipolar n-type semiconductors for high-performance organic electronic devices.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.5c02293.

> Experimental section; Characterization techniques; device fabrication; ^{1}H and ^{13}C NMR spectra of intermediates, monomers, and polymers; GPC images; Theoretical calculations; Cyclic voltammograms; Thermoelectric, UV–vis, TGA, DSC, AFM, and GIWAXS plots (PDF)

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Notes

The authors declare no competing financial interest.

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